# Elementary Analysis

## Chapter 3, Section 20

Limits of Functions
Exercises 20.1–20.20, with every subpart

Mathjock

**Text and numbering.** Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 3. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

**Reading philosophy.** Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

**Conventions.** $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

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# 20 Limits of functions

The approach set is part of a limit's meaning. A two-sided limit at $a$ uses nearby points different from $a$; a right-hand limit uses points strictly greater than $a$. The value at $a$ is irrelevant to these punctured limits. Limits may be finite or may equal one of the two infinite symbols. "Does not exist" means that none of these possibilities is a limit.

The sketches below are original schematic plots in finite windows. Open circles mark missing points. Dashed lines mark asymptotes; the axes themselves also mark an asymptote when appropriate. No finite drawing can show all the oscillations accumulating at zero. The formulas and estimates specify those oscillations exactly.

## 20.1 Exercise 20.1: A sign graph and its five limiting behaviors

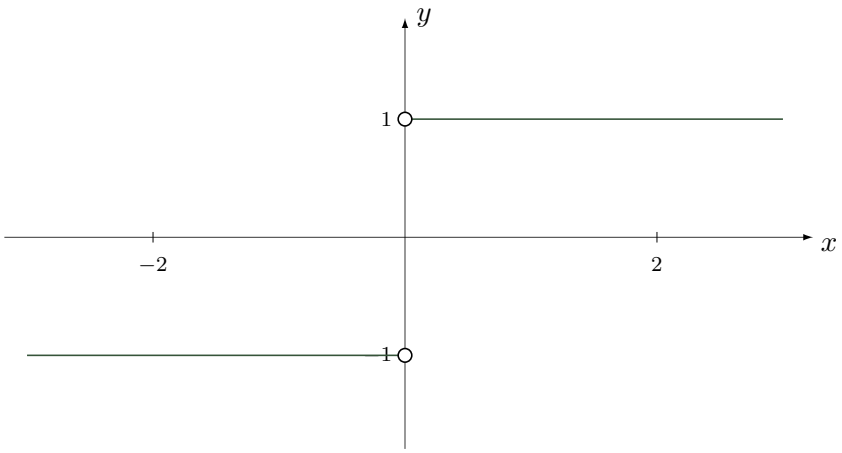
**What we must show.** Sketch $f(x) = x/|x|$ and inspect the five limits requested in the exercise, in their stated order.

**Separate the positive and negative halves of the domain.**

The denominator is zero at $x = 0$, so the domain is $\mathbb{R} \setminus \{0\}$. For $x > 0$, $|x| = x$, whereas for $x < 0$, $|x| = -x$. Consequently

$$f(x) = \begin{cases} 1, & x > 0, \\ -1, & x < 0. \end{cases}$$

The graph consists of two horizontal half-lines, with the endpoints at $(0, 1)$ and $(0, -1)$ omitted.




**Read the limits, keeping the two sides separate.**

| approach | $+\infty$ | $0^+$ | $0^-$ | $-\infty$ | 0 |
|---|---|---|---|---|---|
| limit | 1 | 1 | $-1$ | $-1$ | does not exist |

The right-hand values are constantly 1 and the left-hand values are constantly $-1$. Since these two one-sided limits differ, Theorem 20.10 excludes a two-sided limit. Detailed verification of the assertions is given in Exercise 20.5.

**Conclusion.** The five answers are $1, 1, -1, -1$, and nonexistence.

## 20.2 Exercise 20.2: Two halves of a quadratic graph

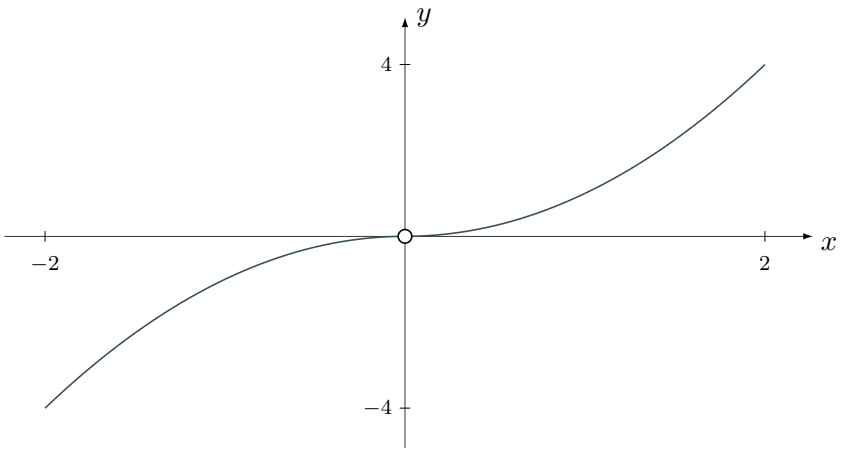
**What we must show.** Repeat the five-limit inspection for $f(x) = x^3/|x|$.

**Cancel only where the denominator is nonzero.**

For $x > 0$, $x^3/|x| = x^2$. For $x < 0$, division by $-x$ gives $x^3/|x| = -x^2$. Thus

$$f(x) = \begin{cases} x^2, & x > 0, \\ -x^2, & x < 0. \end{cases}$$

The right branch rises to $+\infty$, the left branch falls to $-\infty$ as $x$ runs toward the corresponding infinity, and both branches approach the missing point $(0, 0)$.




| approach | $+\infty$ | $0^+$ | $0^-$ | $-\infty$ | $0$ |
|---|---|---|---|---|---|
| limit | $+\infty$ | $0$ | $0$ | $-\infty$ | $0$ |

The two-sided limit is 0 even though $f(0)$ is undefined.

**Conclusion.** The five answers are $+\infty, 0, 0, -\infty, 0$. Exercise 20.6 proves them.

## 20.3 Exercise 20.3: The sine quotient and its removable hole

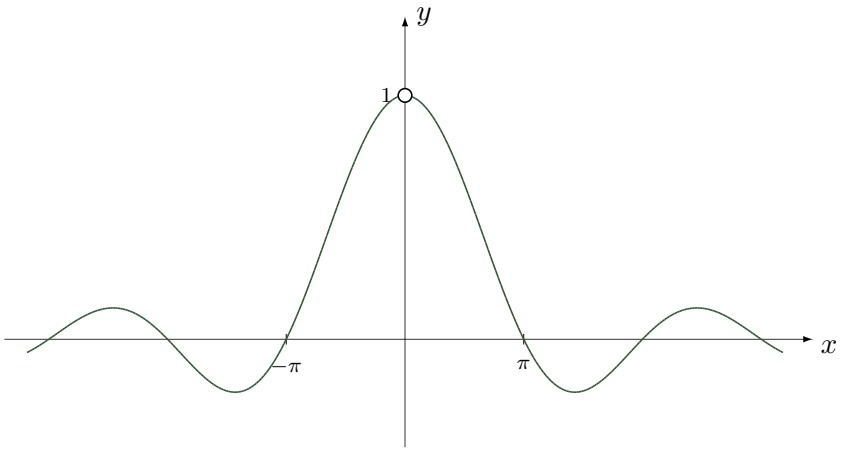
**What we must show.** Sketch $f(x) = \sin x/x$, $x \neq 0$, and inspect the five limits.

**Identify the hole, symmetry, and distant oscillations.**

Since $\sin(-x) = -\sin x$, division by $-x$ shows that $f(-x) = f(x)$. The graph is even. Its zeros are the nonzero integer multiples of $\pi$, and

$$|f(x)| \leq \frac{1}{|x|} \qquad (x \neq 0).$$

The oscillations therefore shrink at either infinity. Near zero, the curve approaches height 1, with $(0, 1)$ omitted.




| approach | $+\infty$ | $0^+$ | $0^-$ | $-\infty$ | 0 |
|---|---|---|---|---|---|
| limit | 0 | 1 | 1 | 0 | 1 |

**Conclusion.** The five limits are $0, 1, 1, 0, 1$. Exercise 20.7 supplies the estimates and the proof near zero.

## 20.4 Exercise 20.4: Oscillations at zero and a horizontal asymptote

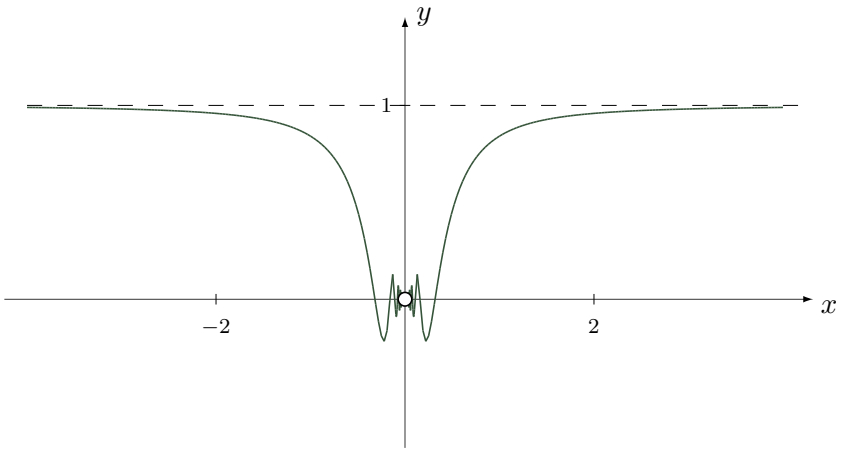
**What we must show.** Repeat the inspection for $f(x) = x\sin(1/x)$, $x \neq 0$.

**Describe both scales of the graph.**

The function is even, since $(-x)\sin(-1/x) = x\sin(1/x)$. Its zeros are $x = 1/(k\pi)$ for nonzero integers $k$. Near zero they accumulate, but

$$|x\sin(1/x)| \leq |x|\,,$$

so the entire oscillating graph is squeezed into a narrowing pair of straight lines. The origin is a hole. At either infinity put $u = 1/x$; then $x\sin(1/x) = \sin u/u$ approaches 1.



The dashed horizontal line is $y = 1$. Only finitely many of the oscillations near the missing origin can be displayed.

| approach | $+\infty$ | $0^+$ | $0^-$ | $-\infty$ | $0$ |
|---|---|---|---|---|---|
| limit | 1 | 0 | 0 | 1 | 0 |

**Conclusion.** The five answers are $1, 0, 0, 1, 0$. Exercise 20.8 proves both the zero and infinity assertions.

## 20.5 Exercise 20.5: Proof of the sign-function limit assertions

**What we must show.** Prove every limit and nonexistence assertion in Exercise 20.1.

**The two positive-side limits.**

For $x > 0$, $f(x) = 1$, so $|f(x) - 1| = 0 < \varepsilon$ for every $\varepsilon > 0$. For the right-hand limit at zero, any $\delta > 0$ works, because $0 < x < \delta$ implies $x > 0$. For the limit at $+\infty$, the threshold $x > 0$ works for every $\varepsilon$. Thus these limits both equal 1.

**The two negative-side limits.**

For $x < 0$, $f(x) = -1$, so $|f(x) - (-1)| = 0$. For the left-hand limit at zero, any $\delta > 0$ works whenever $-\delta < x < 0$. For the limit at $-\infty$, all $x < 0$ already have the required value. Thus these two limits equal $-1$.

**Why zero has no two-sided limit of any kind.**

The one-sided limits are distinct finite numbers. By Theorem 20.10 a two-sided limit, including an infinite limit, would have to agree with both. It cannot. Equivalently, the sequences $x_n = 1/n$ and $y_n = -1/n$ both approach zero, but their function values are constantly 1 and $-1$, respectively.

**Conclusion.** All four one-sided or infinity limits hold, and the two-sided limit at zero does not exist.

## 20.6 Exercise 20.6: Proof of the signed-quadratic limit assertions

**What we must show.** Prove all five assertions in Exercise 20.2.

**The absolute value controls every approach to zero.**

For $x \neq 0$, the piecewise formula gives $|f(x)| = x^2$. Given $\varepsilon > 0$, take $\delta = \sqrt{\varepsilon} > 0$. Then

$$0 < |x| < \delta \quad \Longrightarrow \quad |f(x) - 0| = x^2 < \delta^2 = \varepsilon.$$

This is the two-sided limit 0. The same implication restricted to $x > 0$ or $x < 0$ proves the corresponding one-sided limits.

**The two infinities have different signs.**

Fix $M > 0$. If $x > \sqrt{M}$, then $x > 0$ and $f(x) = x^2 > M$. Hence $f(x) \to +\infty$ as $x \to +\infty$. If $x < -\sqrt{M}$, then $x < 0$, $x^2 > M$, and $f(x) = -x^2 < -M$. This is $f(x) \to -\infty$ as $x \to -\infty$.

**Conclusion.** The three limits at zero are 0; the limits at positive and negative infinity are $+\infty$ and $-\infty$, respectively.

## 20.7 Exercise 20.7: Proof of the sine-quotient limit assertions

**What we must show.** Justify all five limits in Exercise 20.3.

**At either infinity, use a bound that does not depend on phase.**

For $x \neq 0$, $|\sin x/x| \leq 1/|x|$. Given $\varepsilon > 0$, if $|x| > 1/\varepsilon$, then

$$\left|\frac{\sin x}{x} - 0\right| \leq \frac{1}{|x|} < \varepsilon.$$

Taking either $x > 1/\varepsilon$ or $x < -1/\varepsilon$ proves the limits 0 at $+\infty$ and $-\infty$.

**At zero, justify the familiar sine limit.**

We use the Mean Value Theorem, already permitted in Section 19, and the familiar identities $(\sin t)' = \cos t$ and $\sin 0 = 0$. For every nonzero $x$, the theorem on the interval with endpoints 0 and $x$ gives a point $c_x$ strictly between these endpoints such that

$$\frac{\sin x - \sin 0}{x - 0} = \cos c_x, \qquad |c_x| < |x|\,.$$

The hypotheses hold because sine is continuous on that closed interval and differentiable on its interior.

Let $\varepsilon > 0$. Continuity of cosine at zero, and $\cos 0 = 1$, give $\eta > 0$ such that $|t| < \eta$ implies $|\cos t - 1| < \varepsilon$. If $0 < |x| < \eta$, then $|c_x| < \eta$, so

$$\left|\frac{\sin x}{x} - 1\right| = |\cos c_x - 1| < \varepsilon.$$

This proves the two-sided limit 1. Restricting the same estimate to positive or negative $x$ proves both one-sided limits. It also verifies directly the continuous extension discussed in Section 19, Example 9.

**Conclusion.** The two infinity limits are 0 and all three limits at zero equal 1.

## 20.8 Exercise 20.8: Proof for the oscillating product

**What we must show.** Prove the five assertions in Exercise 20.4.

**At zero, the small prefactor controls every oscillation.**

The bound $|\sin t| \leq 1$ gives

$$|x \sin(1/x) - 0| \leq |x| \qquad (x \neq 0).$$

Given $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $0 < |x| < \delta$ forces the displayed error to be less than $\varepsilon$. Hence the two-sided limit is 0, and so are its two one-sided restrictions.

**At infinity, transform the expression rather than its factors separately.**

For $u = 1/x$, one has $u \neq 0$ and

$$x \sin(1/x) = \frac{\sin u}{u}.$$

Exercise 20.7 proved that for every $\varepsilon > 0$ there is $\eta > 0$ with

$$0 < |u| < \eta \quad \Longrightarrow \quad \left|\frac{\sin u}{u} - 1\right| < \varepsilon.$$

If $|x| > 1/\eta$, then $0 < |1/x| < \eta$, so this bound applies. It works for positive and negative $x$, proving a limit of 1 at both infinities. Multiplying the separate informal limits $x \to \infty$ and $\sin(1/x) \to 0$ would not be a valid finite product-limit calculation; the quotient identity supplies the needed control.

**Conclusion.** All three zero limits equal 0 and both infinity limits equal 1.

## 20.9 Exercise 20.9: A reciprocal term and a slant asymptote

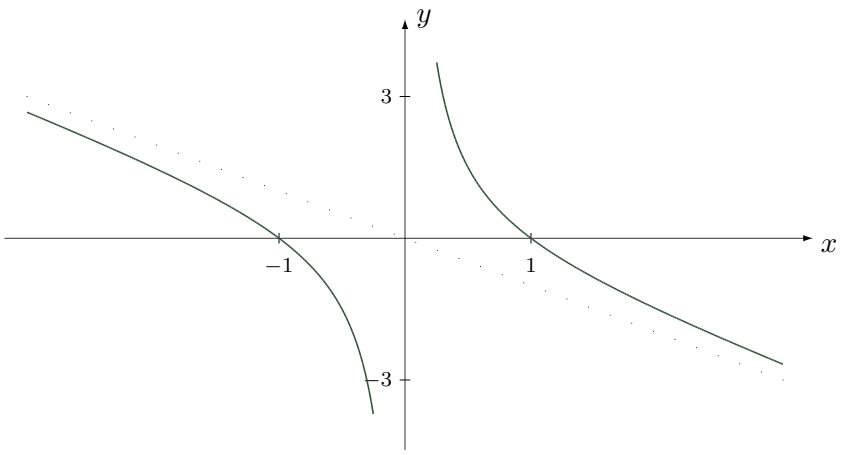
**What we must show.** Sketch $f(x) = (1 - x^2)/x$ and inspect the same five limits.

**Rewrite the function and identify the branches.**

The domain excludes zero, and for $x \neq 0$,

$$f(x) = \frac{1}{x} - x.$$

The graph is odd, since $f(-x) = -f(x)$. It crosses the $x$-axis at $x = \pm 1$. The line $x = 0$ is a vertical asymptote, while $y = -x$ is a slant asymptote because $f(x) - (-x) = 1/x \to 0$ at either infinity.




The dotted diagonal is $y = -x$. The vertical axis is the other asymptote; neither branch contains a point at $x = 0$.

| approach | $+\infty$ | $0^+$ | $0^-$ | $-\infty$ | 0 |
|---|---|---|---|---|---|
| limit | $-\infty$ | $+\infty$ | $-\infty$ | $+\infty$ | does not exist |

**Conclusion.** The answers are $-\infty, +\infty, -\infty, +\infty$, and nonexistence. Exercise 20.10 proves them.

## 20.10 Exercise 20.10: Proof for the reciprocal-minus-linear function

**What we must show.** Prove every limit assertion in Exercise 20.9.

**At positive infinity, the negative linear term dominates.**

Fix $M > 0$, and choose $R = \max\{1, M + 1\}$. If $x > R$, then $1/x < 1$ and $x > M + 1$. Consequently

$$f(x) = \frac{1}{x} - x < 1 - x < -M.$$

This proves the limit $-\infty$ at positive infinity. For $x < -R$, put $t = -x > R$. Oddness gives $f(x) = f(-t) = -f(t) > M$. Hence the limit at negative infinity is $+\infty$.

**At zero, the reciprocal term dominates.**

For the same arbitrary $M > 0$, choose $\delta = \min\{1, 1/(M+1)\} > 0$. If $0 < x < \delta$, then $1/x > M + 1$ and $x < 1$, giving

$$f(x) = \frac{1}{x} - x > (M+1) - 1 = M.$$

This proves the right-hand limit $+\infty$ at zero. If $-\delta < x < 0$, put $t = -x \in (0, \delta)$. Then $f(x) = -f(t) < -M$, proving the left-hand limit $-\infty$. The one-sided limits differ, so Theorem 20.10 shows that the two-sided limit does not exist, even as an infinite limit.

**Conclusion.** All five assertions follow from explicit threshold estimates, with the signs preserved on each side of zero.

## 20.11 Exercise 20.11: Three removable algebraic quotients

**What we must show.** Find the three limits by simplifying only at points where the original quotients are defined.

**(a) Factor the difference of squares.**

For $x \neq a$,

$$\frac{x^2 - a^2}{x - a} = \frac{(x-a)(x+a)}{x-a} = x + a.$$

The expression on the right is continuous at $a$. Since the punctured expressions agree, their limits agree, and the answer is $a + a = 2a$. No value of the original quotient at $a$ is needed.

**(b) Multiply by the conjugate, with the domain checked.**

Here $b > 0$. For example, $|x - b| < b/2$ ensures $x > b/2 > 0$, so both square roots exist. If also $x \neq b$, then

$$\frac{\sqrt{x} - \sqrt{b}}{x - b} = \frac{(\sqrt{x} - \sqrt{b})(\sqrt{x} + \sqrt{b})}{(x-b)(\sqrt{x} + \sqrt{b})} = \frac{1}{\sqrt{x} + \sqrt{b}}.$$

The conjugate is positive, and the cancellation uses $x - b \neq 0$. As $x \to b$, continuity of the square-root function gives a denominator tending to $2\sqrt{b} > 0$. The reciprocal limit is therefore $1/(2\sqrt{b})$.

**(c) Factor the difference of cubes.**

The identity $x^3 - a^3 = (x-a)(x^2 + ax + a^2)$ follows by multiplying out; the two mixed terms cancel. For $x \neq a$ it gives

$$\frac{x^3 - a^3}{x - a} = x^2 + ax + a^2.$$

This polynomial is continuous at $a$, so the limit is $a^2 + a^2 + a^2 = 3a^2$.

**Conclusion.** The limits in (a), (b), and (c) are $2a$, $1/(2\sqrt{b})$, and $3a^2$, respectively.

## 20.12 Exercise 20.12: One simple pole and one squared pole

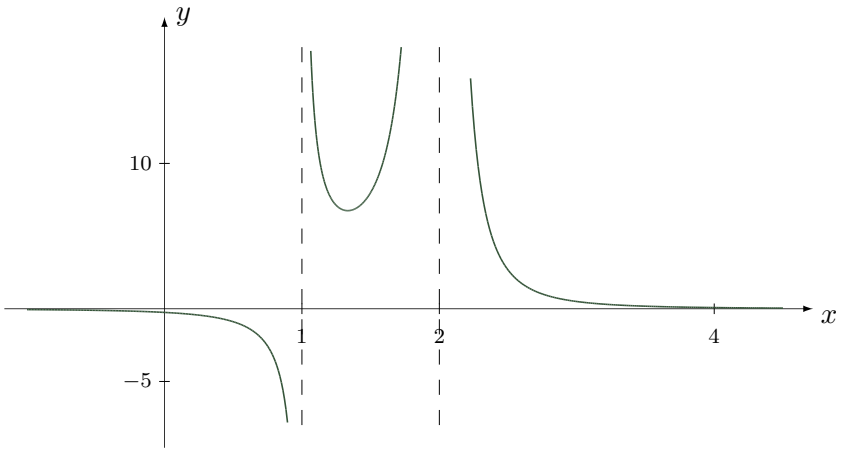
**What we must show.** Sketch $f(x) = 1/[(x-1)(x-2)^2]$, determine all four requested one-sided limits, and decide the two two-sided limits.

**(a) Use signs and poles to organize the sketch.**

The domain is $\mathbb{R} \setminus \{1, 2\}$. For $x \neq 2$, $(x-2)^2 > 0$, so the sign of $f(x)$ is the sign of $x - 1$. The graph is negative to the left of 1 and positive on both $(1, 2)$ and $(2, \infty)$. There are vertical asymptotes at 1 and 2 and a horizontal asymptote $y = 0$ at both infinities. For additional shape information, differentiation on each domain interval gives

$$f'(x) = -\frac{3x-4}{(x-1)^2(x-2)^3}.$$

Thus the middle positive branch decreases on $(1, 4/3)$ and increases on $(4/3, 2)$, reaching $f(4/3) = 27/4$. The left branch decreases, and the branch to the right of 2 decreases to zero. These derivative facts concern only the sketch; the limit estimates below do not require them.




The dashed vertical lines are $x = 1$ and $x = 2$. The portions outside the finite plotting window continue toward the indicated infinite limits.

**(b), near 2: the squared factor is positive on both sides.**

If $0 < |x-2| < 1/2$, then $1/2 < x - 1 < 3/2$. Hence

$$f(x) = \frac{1}{(x-1)(x-2)^2} > \frac{2}{3\,|x-2|^2} > 0.$$

Given $M > 0$, choose $\delta = \min\{1/2, \sqrt{2/(3M)}\}$. Then $0 < |x-2| < \delta$ makes this last lower bound exceed $M$. Thus both $\lim_{x\to 2^+} f(x)$ and $\lim_{x\to 2^-} f(x)$ equal $+\infty$.

**(b), near 1: the unsquared factor changes sign.**

If $0 < |x-1| < 1/2$, then $1/2 < |x-2| < 3/2$, so

$$|f(x)| = \frac{1}{|x-1|\,(x-2)^2} > \frac{4}{9\,|x-1|}.$$

For $M > 0$, taking $\delta = \min\{1/2, 4/(9M)\}$ makes the magnitude exceed $M$. For $x > 1$ the function is positive, so the right-hand limit is $+\infty$. For $x < 1$ it is negative, so the left-hand limit is $-\infty$.

**(c) Compare the two sides at each pole.**

At 2 the two sides agree, giving $\lim_{x\to 2} f(x) = +\infty$ by Theorem 20.10. At 1 they disagree, so $\lim_{x\to 1} f(x)$ does not exist.

**Conclusion.** In the requested order the one-sided limits are $+\infty, +\infty, +\infty, -\infty$. The two-sided limit at 2 is $+\infty$; the one at 1 does not exist.

## 20.13 Exercise 20.13: Finite limit laws with reciprocal and square root

**What we must show.** From $\lim_{x\to a} f(x) = 3$ and $\lim_{x\to a} g(x) = 2$, prove all three asserted limits.

**(a) Apply the finite sum and product laws.**

The limits are finite, so Theorem 20.4 gives

$$\lim_{x\to a} g(x)^2 = 2 \cdot 2 = 4, \qquad \lim_{x\to a} 3f(x) = 3 \cdot 3 = 9.$$

Adding these limits gives

$$\lim_{x\to a}[3f(x) + g(x)^2] = 9 + 4 = 13.$$

**(b) Ensure the reciprocal is defined before using its law.**

Since $g(x) \to 2$, there is a punctured neighborhood of $a$ on which $|g(x) - 2| < 1$. There $g(x) > 1$, so $g(x) \neq 0$. On that neighborhood the quotient law applies, with the constant numerator 1 and denominator limit $2 \neq 0$. Therefore

$$\lim_{x\to a} \frac{1}{g(x)} = \frac{1}{2}.$$

**(c) Check the square-root domain near the limiting point.**

The sum and scalar laws first give

$$h(x) = 3f(x) + 8g(x) \longrightarrow 3 \cdot 3 + 8 \cdot 2 = 25.$$

For all sufficiently close $x \neq a$, one has $|h(x) - 25| < 1$, hence $h(x) > 24 > 0$. Thus the square root is defined throughout a punctured neighborhood. The square-root function is continuous at 25; Theorem 20.5, applied on this neighborhood, gives

$$\lim_{x\to a} \sqrt{3f(x) + 8g(x)} = \sqrt{25} = 5.$$

**Conclusion.** The three limits are 13, 1/2, and 5. Both nonzero-denominator and nonnegative-radicand conditions hold on a sufficiently small punctured neighborhood.

## 20.14 Exercise 20.14: The two one-sided infinite reciprocal limits

**What we must show.** Prove $\lim_{x\to 0^+} 1/x = +\infty$ and $\lim_{x\to 0^-} 1/x = -\infty$ directly.

**Right side: choose the distance from the target height.**

Fix $M > 0$ and put $\delta = 1/M > 0$. If $0 < x < \delta$, reciprocation of positive numbers reverses the strict inequality and yields $1/x > 1/\delta = M$. This is precisely the threshold definition of the first limit.

**Left side: keep track of the negative sign.**

For the same $M$ and $\delta$, if $-\delta < x < 0$, then $0 < -x < \delta$. The previous positive estimate gives $1/(-x) > M$. Multiplying by $-1$ reverses the inequality: $1/x < -M$. Since every negative threshold is $-M$ for some $M > 0$, this proves the second limit.

**Conclusion.** The reciprocal tends to $+\infty$ from the right and to $-\infty$ from the left.

## 20.15 Exercise 20.15: Two limits of the shifted reciprocal cube

**What we must show.** For the function in Example 4, $f(x) = 1/(x-2)^3$ with $x \neq 2$, prove the limits at $-\infty$ and $2^+$ requested in the exercise.

**At negative infinity, bound the absolute error from zero.**

Let $\varepsilon > 0$ and set $A = -\varepsilon^{-1/3} < 0$. If $x < A$, then $x < 0$ and $|x-2| = 2 - x > -x > \varepsilon^{-1/3}$. Cubing positive quantities and then reciprocating gives

$$|f(x) - 0| = \frac{1}{|x-2|^3} < \varepsilon.$$

This proves $\lim_{x\to-\infty} f(x) = 0$.

**At the pole from the right, choose a small positive denominator.**

Given $M > 0$, put $\delta = M^{-1/3} > 0$. If $2 < x < 2 + \delta$, then $0 < x - 2 < \delta$. Hence $(x-2)^3 < \delta^3 = 1/M$ and $f(x) = 1/(x-2)^3 > M$. Thus $\lim_{x\to 2^+} f(x) = +\infty$.

**Conclusion.** The two limits are 0 and $+\infty$, respectively.

## 20.16 Exercise 20.16: Order of right-hand limits and the loss of strictness

**What we must show.** Prove the order assertion in part (a), and determine whether strict pointwise order implies strict order of limits in part (b). The existing limits may be finite or infinite.

**(a) Approach through one common sequence.**

Shrink $b > a$ if necessary so that the interval $(a, b)$ lies in the neighborhoods used to define both right-hand limits. Let

$$x_n = a + \frac{b-a}{n+1}.$$

Then $a < x_n < b$ and $x_n \to a$. The hypotheses therefore give

$$f_1(x_n) \longrightarrow L_1, \qquad f_2(x_n) \longrightarrow L_2, \qquad f_1(x_n) \le f_2(x_n) \quad (n \in \mathbb{N}).$$

The order rule for existing sequence limits, Exercise 9.9(c), now gives $L_1 \le L_2$, including the possible infinite values. To recall why that order rule includes these values, if $L_1 > L_2$, choose a finite real $c$ strictly between them. Convergence would eventually give both $f_1(x_n) > c$ and $f_2(x_n) < c$, contrary to the pointwise inequality.

**(b) Give a counterexample with strictly ordered values.**

On $(a, a+1)$ put $f_1(x) = 0$ and $f_2(x) = x - a$. Every point satisfies $f_1(x) < f_2(x)$, since $x > a$. Nevertheless both right-hand limits at $a$ equal 0. The positive gap $x - a$ shrinks to zero, so no strict gap remains between the limits.

**Conclusion.** Part (a) gives $L_1 \le L_2$. The answer to part (b) is no: equality of limits is possible despite strict inequality at every point.

## 20.17 Exercise 20.17: The squeeze lemma proves existence of the middle limit

**What we must show.** Suppose $f_1(x) \le f_2(x) \le f_3(x)$ on $(a, b)$, and both outside functions have right-hand limit $L$ at $a$. Prove that $f_2$ has that same limit, without assuming it exists.

**Finite $L$: enclose all middle values in a prescribed error interval.**

Let $\varepsilon > 0$. The two given limits supply $\delta_1, \delta_3 > 0$ such that

$$\begin{aligned} a < x < a + \delta_1 \quad &\Longrightarrow \quad L - \varepsilon < f_1(x) < L + \varepsilon, \\ a < x < a + \delta_3 \quad &\Longrightarrow \quad L - \varepsilon < f_3(x) < L + \varepsilon. \end{aligned}$$

Take $\delta = \min\{\delta_1, \delta_3, b - a\} > 0$. Whenever $a < x < a + \delta$, the point lies in $(a, b)$ and

$$L - \varepsilon < f_1(x) \le f_2(x) \le f_3(x) < L + \varepsilon.$$

Thus $|f_2(x) - L| < \varepsilon$. This verifies the definition of its limit and establishes existence as part of the conclusion.

**Infinite $L$: one of the outside functions already forces the threshold.**

If $L = +\infty$, let $M > 0$. The limit of $f_1$ gives $\delta_1 > 0$ such that $a < x < a + \delta_1$ implies $f_1(x) > M$. Restrict also to $x < b$. Then $f_2(x) \ge f_1(x) > M$, proving $f_2(x) \to +\infty$. If $L = -\infty$, let $M > 0$ and use the upper function instead: eventually $f_3(x) < -M$, so $f_2(x) \le f_3(x) < -M$. This proves $f_2(x) \to -\infty$.

**Conclusion.** The middle limit exists and equals $L$ in every finite or infinite case. No order theorem requiring its prior existence was used.

## 20.18 Exercise 20.18: Rationalizing a square-root difference

**What we must show.** Show the limit at zero exists for

$$f(x) = \frac{\sqrt{1 + 3x^2} - 1}{x^2} \qquad (x \neq 0),$$

and determine its value.

**Check every factor used in the rationalization.**

For every real $x$, $1+3x^2 \geq 1$, so the square root exists and $\sqrt{1+3x^2}+1 \geq 2 > 0$. If $x \neq 0$, then $x^2 > 0$. We may therefore multiply the numerator and denominator by the conjugate and cancel the nonzero factor $x^2$:

$$\begin{aligned} f(x) &= \frac{(\sqrt{1+3x^2}-1)(\sqrt{1+3x^2}+1)}{x^2(\sqrt{1+3x^2}+1)} \\ &= \frac{(1+3x^2)-1}{x^2(\sqrt{1+3x^2}+1)} = \frac{3}{\sqrt{1+3x^2}+1}. \end{aligned}$$

**Use a continuous expression that also makes sense at zero.**

The polynomial $1+3x^2$ tends to 1 as $x \to 0$. Continuity of square root gives a denominator tending to $\sqrt{1}+1 = 2 \neq 0$. The reciprocal and scalar limit laws yield

$$\lim_{x\to 0} f(x) = \frac{3}{2}.$$

This establishes existence, not just a possible value: the original function agrees on every nonzero point with an expression continuous at zero.

**Conclusion.** The limit exists and equals 3/2.

## 20.19 Exercise 20.19: Changing the right-hand approach interval

**What we must show.** Let $a < b_1 < b_2$ and let $f$ be defined on $(a, b_2)$. Show that existence and value of the limit along $(a, b_i)$ are independent of choosing $i = 1$ or $i = 2$.

**A limit along the larger interval restricts to the smaller one.**

Suppose the limit along $S_2 = (a, b_2)$ exists and equals $L$. Every sequence in $S_1 = (a, b_1)$ is also a sequence in $S_2$, since $S_1 \subseteq S_2$. Thus whenever a sequence in $S_1$ tends to $a$, its function values tend to $L$ by the given limit. This is the sequential definition of the limit along $S_1$, with the same value.

**A sequence in the larger interval eventually enters the smaller one.**

Conversely, suppose the limit along $S_1$ exists and equals $L$. Take an arbitrary sequence $(x_n)$ in $S_2$ tending to $a$. The positive number $b_1 - a$ is an allowable tolerance for convergence. Therefore for some integer $N$,

$$n > N \quad \Longrightarrow \quad |x_n - a| < b_1 - a.$$

Since all these $x_n > a$, the inequality says $a < x_n < b_1$. Hence the tail $x_{N+1}, x_{N+2}, \ldots$ is a sequence in $S_1$ and still tends to $a$. By the assumed smaller-interval limit, the corresponding function values tend to $L$. Adding back finitely many initial terms does not change a finite or infinite sequence limit. Thus $f(x_n) \to L$ for the whole arbitrary sequence, proving the limit along $S_2$.

Both approach sets admit sequences tending to $a$, for example $a + (b_1 - a)/(n+1)$, so their limits are meaningful under Definition 20.1 and are unique. The proof did not require $L$ to be finite.

**Conclusion.** Either interval defines the same right-hand limit, finite or infinite, which is denoted $\lim_{x\to a^+} f(x)$.

## 20.20 Exercise 20.20: Infinite sum and product limit rules

**What we must show.** Assume $\lim_{x\to a^S} f_1(x) = +\infty$ and that $L_2 = \lim_{x\to a^S} f_2(x)$ exists. Prove all three asserted rules and analyze the zero case in part (d). Here $a$ may be finite or infinite, with the approach set $S$ as in Definition 20.1.

**Use an arbitrary admissible sequence, so all approach types are covered.**

Take any sequence $x_n \in S$ with $x_n \to a$, and write $u_n = f_1(x_n)$, $v_n = f_2(x_n)$. The hypotheses give $u_n \to +\infty$ and $v_n \to L_2$. We prove each conclusion for this arbitrary sequence. Definition 20.1 will then give the corresponding function limit.

**(a) A finite or positive-infinite second limit cannot cancel the first.**

Suppose $L_2 \neq -\infty$. There is a finite real lower bound $c$ for all sufficiently late $v_n$: if $L_2$ is finite, take $c = L_2 - 1$; if $L_2 = +\infty$, take $c = 0$. For any $M > 0$, the limit $u_n \to +\infty$ eventually gives $u_n > M - c$ (or a larger positive threshold if needed). On the common tail,

$$u_n + v_n \geq u_n + c > M.$$

Hence $u_n + v_n \to +\infty$, so $\lim_{x\to a^S}(f_1 + f_2)(x) = +\infty$.

**(b) A positive second limit gives a fixed positive factor.**

Suppose $0 < L_2 \leq +\infty$. If $L_2$ is finite, take $c = L_2/2 > 0$. If $L_2 = +\infty$, take $c = 1$. In either case $v_n \geq c$ eventually. For any $M > 0$, eventually $u_n > M/c > 0$ as well. Multiplication by this positive $u_n$ preserves the inequality, giving

$$u_n v_n \geq c u_n > M.$$

Thus $u_n v_n \to +\infty$, proving the asserted product rule.

**(c) Reflect a negative second limit before applying the positive rule.**

Suppose $-\infty \leq L_2 < 0$. Then $-v_n \to -L_2 > 0$, where $-L_2 = +\infty$ if $L_2 = -\infty$. Part (b)'s argument applied to $u_n$ and $-v_n$ gives $u_n(-v_n) \to +\infty$. Negating reverses every threshold inequality, so $u_n v_n \to -\infty$. Since the admissible sequence was arbitrary, this proves $\lim_{x\to a^S}(f_1 f_2)(x) = -\infty$.

**(d) A zero second limit does not determine the product.**

Use the common approach $x \to 0^+$ on $S = (0, 1)$, and fix $f_1(x) = 1/x^2$, which tends to $+\infty$. For any prescribed real $c$, choose $f_2(x) = cx^2$. Then $f_2(x) \to 0$ but $f_1(x)f_2(x) = c$, so any finite real limit can occur. If $f_2(x) = x$, its limit is again 0 but the product is $1/x \to +\infty$. If $f_2(x) = -x$, the product tends to $-\infty$.

There need not be any product limit: choose $f_2(x) = x^2 \sin(1/x)$. The bound $|f_2(x)| \leq x^2$ gives $f_2(x) \to 0$, while the product is $\sin(1/x)$. Along

$$x_n = \frac{1}{2\pi n + \pi/2}, \qquad y_n = \frac{1}{2\pi n + 3\pi/2},$$

both of which belong to $(0, 1)$ and tend to zero, its values are 1 and $-1$, respectively. These incompatible sequence limits exclude a function limit.

**Conclusion.** Parts (a), (b), and (c) give $+\infty$, $+\infty$, and $-\infty$, respectively. When $L_2 = 0$, the product may have any finite limit, either infinite limit, or no limit.